Notes about recovering data from HDs

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Introduction

Memory devices, such as hard drives, SSDs, USB drives, and other forms of digital storage, store vast amounts of data in various formats. Retrieving relevant information from these devices can be a challenging task, especially when the nature of the information sought is not well-defined or when the dataset is extremely large. As a consequence, we need to cope with different issues, listed in the following.

2. The number of possible files and applications can be huge

Modern memory devices can store millions of files, ranging from documents and images to system logs, databases, and application files. Additionally, the presence of diverse applications adds complexity, since they may store data in unique formats or not-so-easy-to-reach locations within the device. Consequently, the investigator or data specialist may face challenges such as:

- navigating the file system architecture to locate hidden or obscured files.
- reading a wide range of file formats, some of which may be proprietary or encrypted.
- Identifying data's origin (user-generated data, system files, or application-specific files).

3. Look only for what is relevant (but exhaustive analysis could be needed anyway)

One of the primary challenges in retrieving relevant data is filtering out the irrelevant information while ensuring that no critical data is missed: in many cases, the relevance depeds on the specific context.

For example:

- In a digital forensics cases, relevant information might include email records, browsing history, deleted files, or metadata associated with documents.
- In a data recovery, the focus may be on restoring lost files, which might require scanning through corrupted or fragmented data structures.
- Target specific types of data could be needed such as financial records or customer information.

It could be necessary to perform an exhaustive analysis, i.e., scanning and indexing the entire memory device, searching for keywords, file signatures, or other markers that indicate the presence of relevant data (time-consuming and resource-intensive).

4. Different HD and OS tools could be needed

The availability of tools for disk and operating system (OS) investigation spans a wide spectrum and varies depeding on the characteristics of the platform and licensing. We can find commercial, free, and open source solutions.

- **Commercial tools**: These are proprietary and often feature-rich solutions developed by companies specializing in data forensics and recovery (e.g., EnCase, FTK, X-Ways). These tools typically offer extensive support, frequent updates, and powerful

analysis capabilities, making them suitable for professional use in law enforcement or corporate environments.

- **free tools**: Some companies or developers provide free versions of their softwares with limited features, allowing basic investigation and data retrieval tasks. These can be valuable for smaller organizations or individual investigators working on a budget.

- **open source solutions**: These are developed by the community and provide transparency, allowing users to view and modify the source code (e.g., Autopsy, Sleuth Kit, TestDisk). Open source tools often excel in flexibility and adaptability, offering various plug-ins or modules that allow customization to meet specific requirements.

Each category of tool offers distinct advantages and trade-offs. Commercial tools provide advanced capabilities but may be costly, while free and open-source alternatives may require more technical expertise but offer greater adaptability.

5. Operations and procedures must be accurately chosen

Accurate planning and execution are essential in memory device investigations to ensure the correct data is retrieved efficiently. The choice of operations depends on various factors, including:

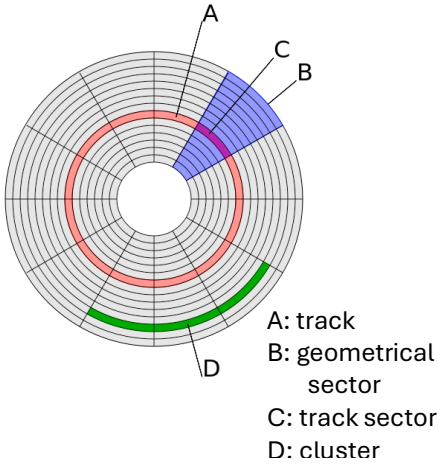
- **type of memory device**: For instance, hard disk drives (HDDs) and solid-state drives (SSDs) function differently, with SSDs having wear-leveling algorithms that complicate data recovery processes. Other devices like USB drives or network-attached storage (NAS) have their own protocols and storage mechanisms.
- **operating system**: Different OS environments (e.g., Windows, Linux, macOS) use distinct file systems (e.g., NTFS, ext4, APFS). This affects how files are stored, deleted, and recovered, and investigators must choose tools compatible with the OS in question.
- **nature of the data**: If the goal is to recover deleted files, the approach may involve scanning file system metadata and unallocated space for remnants of previously stored data. If the aim is to extract application-specific data, like logs or database entries, specialized knowledge of the application's storage practices is required.

How traditional hard disks store data: a quick overview

In computer file systems, a **cluster** (or allocation unit) is the fundamental unit of disk space allocation used for managing files and directories.

Indeed, to optimize disk space utilization and management, filesystems group multiple sectors together into clusters, rather than allocating individual sectors for each file. This approach allows for more efficient data handling, especially when dealing with large volumes of information.

Thus, while every sector *can be assigned to a single file*, a single file can be *stored in multiple sectors*.



Clusters can either be

allocated—meaning they are currently in use to store data, metadata, or programs

non-allocated, indicating they are available and ready to be used for new data storage.

Understanding the state of these clusters is essential in disk analysis, as it enables investigators to identify and recover both active and deleted files, enhancing the efficiency and accuracy of data retrieval efforts.

As a matter of fact, information retrieval and analysis must focus on **slack space.** This term refers to the unused space between the end of a file and the end of the last assigned sector within a cluster.
This space is considered wasted since it is allocated but not used for storing actual file data (see image below).



Example: NTFS
Clusters: 4Kb
File: 9 Kb = 3 clusters for a

However, this unused area can still contain remnants of previously deleted files or fragments of other data, making it valuable in digital forensics for recovering hidden or deleted information (see the example below).

Example 2: NTFS
Clusters: 4Kb File A: 12 Kb = 3 clusters for a total of 12 Kb
No slack space.
Remove File A and write File B: part of the data are still present!



The term **slack space** can also be more broadly applied to any part of the memory that is no longer indexed or actively managed by the file system, potentially holding traces of prior data. It's important to note that while each sector is assigned exclusively to a single file, a single file may span multiple sectors, depending on its size.
Some data can be retrieved from the so-called **G-list:** damaged sector that can not be used any more but that can be read!

45 % of data that can be read from a hard disk are in slack and non-index spaces!

What about Solid State Drives (SSDs)?

SD cards store data on flash memory chips, which allows for simultaneous access without the delays associated with spinning disks found in traditional hard drives. This design provides faster and more efficient data retrieval. However, recovering files from SD cards and similar flash-based devices can be more challenging compared to traditional storage media. The underlying issue with SSDs and flash memory is that **data blocks must be erased before they can be rewritten**, a process managed by the device's firmware.

This procedure, along with features like **wear leveling** and **garbage collection**, complicates file recovery efforts as these mechanisms may overwrite or eliminate data remnants, making it difficult to recover deleted files.

Let's see in detail the two main commends/operations that are responsible for this.

TRIM (TRIM in ATA set or UNMAP in SCSI)

TRIM is a command used in SSDs (Solid State Drives) to optimize their performance and longevity. It is issued by OS to SSD controller at file deletion, disk formatting, or partition removal in order to signal which sectors can be overwritten (reused).

When files are deleted on an SSD, the operating system marks the space as *available,* but unlike traditional hard drives, SSDs must **erase data blocks before** they can be rewritten. In order to speed-up this process, the TRIM commands **proactively erases the blocks** associated with deleted files when the drive is idle, making the space immediately available for future writes. Without this, the process would significantly *slow down write operations*, as the SSD would have to erase blocks during each write operation.

While TRIM improves speed and efficiency, it poses challenges for data recovery.

Garbage collection

Over time, as files are created, modified, and deleted, the SSD's flash memory can become fragmented, with partially filled or outdated blocks scattered across the drive.
The garbage collection process helps maintain the SSD's efficiency by **recollecting valid data from multiple blocks into a new one** and **then purging the unused old blocks** to free up space for future writes.
This process runs automatically in the background, improving the drive's performance by reducing write times.
Note that it is Implemented on disk itself (not controlled by OS). When PC is shut down, the process could restart when disk is powered on or even if attached to a write blocker.

Note: garbage collection does not occur if the TRIM command has not been issued.

Data carving for SSD devices presents significant challenges, particularly due to the prevalent use of **encryption** and **compression** techniques applied to stored data.

Encryption
Encryption ensures that files are securely stored, making it nearly impossible to recover usable information without the appropriate decryption keys. Even if fragmented or deleted files are identified during the carving process, the data remains inaccessible without this critical information.

Compression
The compression of data complicates data carving because it reduces the size of files by altering their internal structure. Files may not be stored contiguously (fragmented), thus making the reconstruction of a compressed file impossible since decoding operations could be missing important information (contexts, …) stored in other sectors that are not available.

Solutions to recover data from SSDs

While the purging of data blocks is generally expected to occur after the TRIM command is issued, there are specific scenarios and limitations that may prevent this process from functioning effectively. In the following, we report a list of such limits:

- **TRIM command could not be supported or enabled** in certain operating systems. If the OS does not recognize or implement the TRIM command, information could be prevented from being erased and the searched information could be still there.
- **TRIM not working on most RAID, external SSD and NAS**. The functionality of TRIM is limited or disabled for various storage configurations, such as RAID (Redundant Array of Independent Disks), external SSDs, and Network Attached Storage (NAS) systems. AS an example, in RAID setups TRIM command should be communicated to each individual SSD.
- **TRIM is only supported in SATA, eSATA, and SCSI.** The TRIM command is primarily supported through specific interfaces, including SATA (Serial ATA), eSATA, and SCSI (Small Computer System Interface). SSDs that are connected via USB do not benefit from TRIM functionality (unaffected)
- **Firmware Issues.** Additionally, many SSDs have been released with buggy firmware, which can disable the effects of the TRIM command and garbage collection processes. These firmware issues can lead to inefficient data management and performance degradation over time. Users may experience difficulties in utilizing the full capabilities of their SSDs due to these inherent software flaws.
- **Corrupted Files**: a file may become corrupted rather than deleted, and therefore, the data may still be present on the SSD, and the TRIM command may not effectively purge them. This situation presents an opportunity for data recovery efforts.

So what are the best strategies to take?

1. Attach SSD to a secondary channel of the motherboard (not primary drive)
This solution can sometimes bypass issues related to the operating system's management of the primary drive, particularly if the primary drive has encountered errors or corruption. By connecting the SSD to a secondary channel, data recovery software may be able to access the drive more effectively without interference from the operating system. This method can also prevent the SSD from engaging in its usual management processes, such as TRIM and garbage collection, which might otherwise erase recoverable data during a recovery attempt.

2. Checking if the disk is close to full
SSDs use a process called **wear leveling**, which distributes writes evenly across memory blocks to prolong the drive's lifespan. If the drive is nearly full, there may be little room for new writes, leading to a higher chance that the data has not been overwritten. This situation increases the likelihood that deleted files can still be recovered.

3. Checking the characteristics of garbage collectors
The effectiveness and behavior of garbage collection can vary significantly between different SSD manufacturers and models. Understanding the specific garbage collection algorithms used by the SSD can inform recovery strategies, as some vendors implement aggressive garbage collection that may permanently erase deleted data more quickly than others. Knowledge of these characteristics can help recovery professionals estimate the likelihood of successfully retrieving deleted files and tailor their methods accordingly.

Data carving on HDs and SSDs for different filesystems

We already mentioned the fact that, when you delete a file, the system does not immediately remove the actual data from the hard drive. Instead, it erases the address in the table that contains the indexes of all the parts of the file. This means that, even if the file is no longer visible in the normal file list, the data are still there on the drive. Sometimes, deleted files are moved to a special folder called the trash bin, allowing you to recover them easily if you need to.

The data from a deleted file remains on the hard drive until certain conditions occur.
First, if new data is written over the space where the deleted file was stored, it can permanently erase the old data, making recovery impossible. This eventuality becomes more and more likely as the HD becomes full, and therefore, the probability of non-indexed space being use becomes higher.
Second, if a wiping software is used, it intentionally removes data more thoroughly than a simple delete command, ensuring that the information is completely gone.

Note also that data carving can be carried on on some RAM dump.

These considerations alllows us to state that such operations are strongly dependent on how files are stored on hard drives, i.e., on the way filessystems organize data.

To effectively recover a file using data carving, certain **base conditions** must be met:

1. **Header and footer**: the header is the beginning of the file, while the footer is the end, and present characteristic pattern that identify the file type. Indeed, different formats present different prefixes and suffixes. A **magic number** is a constant value used in computing to identify the file type by checking specific sequences of bytes at the beginning and end of a file.

   Examples:
   PDF: starts with %PDF and ends with %EOF
   JPG: starts with 0xFFD8 and ends with 0xFFD9

   These unique identifiers allow software applications to determine the format of a file and ensure that it is processed correctly. By examining these magic numbers, data recovery tools and forensic analysts can quickly ascertain the types of files present on a storage device and their sizes, facilitating effective data carving and recovery efforts.

2. **File fragmentation**: the file must not be fragmented. This means that all parts of the file should be stored together, without being split into different sections on the storage device. If a file is fragmented, it becomes more challenging to piece it back together.

3. **File compression**: compressed files have been altered to save space, making it difficult to recognize the original structure of the file.

4. **File structure**: the file should be located between the header and footer.

While base conditions are important, there are also **advanced conditions** that can affect data recovery:

1. **Non-sequential fragments**: sometimes, the fragments of a file may not be stored in order. This means that parts of the file could be scattered across different locations on the storage device. Data carving can still recover the file if enough fragments are found.

2. **Non-ordered fragments**: fragments might not just be out of sequence but could also be mixed with fragments from other files. This adds complexity to the recovery process, as the carving software needs to sort through.

3. **Missing fragments**: If some fragments are missing, the recovery process becomes even more challenging. However, data carving techniques can sometimes still retrieve useful information.

Considerations
Data carving can be performed directly on the storage device or on a copy of the data, i.e., image. This approach helps to preserve the original data while attempting recovery.

Pay attention that you must be cautious of **false positives**. This occurs when the carving process identifies fragments that appear to be part of a file but do not actually contain valid data.

Overcoming the effects of encryption

A short overview on disk encryption

Disk encryption is a security measure that protects sensitive data stored on hard drives by converting it into a coded format that can only be accessed with the right key or password. This report will explore three types of disk encryption: hardware (HW) encryption, software (SW) encryption, and hybrid encryption, explaining their features and differences in simple terms.

Software Encryption

Software encryption is a method that relies on specific applications to encrypt data. One of the main advantages of software encryption is that it can be used on various machines since it is not tied to a specific hardware platform. Instead, the encription/decription process is tied to a set of keys that can be shared.
However, this flexibility comes at a cost: software encryption tends to be slower than other methods because it uses the computer's processing power to perform the encryption.
Examples of software encryption:

1. **VeraCrypt**: free, open-source disk encryption software that allows users to create a virtual encrypted disk or encrypt entire partitions.
2. **TrueCrypt**: one of the first popular open-source disk encryption tools (no longer actively maintained).
3. **AxCrypt**: simple file encryption software that allows users to encrypt individual files with ease. It integrates with Windows Explorer for quick access and can be used for both personal and business purposes.
4. **BitLocker**: A built-in disk encryption feature in Microsoft Windows that uses AES encryption. It provides full disk encryption and is user-friendly, requiring minimal setup.

5. **GnuPG (GPG)**: A free implementation of the OpenPGP standard, GnuPG allows users to encrypt and sign their data and communications. It is widely used for securing email and files.

Hardware Encryption

In contrast, hardware encryption operates at a firmware level, meaning it is built directly into the hard drive's control system. Many modern hard drives and SSDs come with built-in hardware encryption capabilities (Self-Encrypting Drives or SEDs).
Examples include:

- **Samsung T7 SSD**: Offers hardware encryption with AES 256-bit security.
- **Western Digital My Passport**: A portable drive that includes hardware encryption options.

This method often relies on a component called the **Trusted Platform Module (TPM)** chip, which enhances security by managing encryption keys. Hardware encryption is faster than software encryption because the hard drive's control system handles the encryption process, allowing for quick access to encrypted data. Additionally, hardware encryption encrypts the entire drive, including the boot sector, operating system, temporary files, and swap files, providing comprehensive protection for all data stored on the drive.
TPM chips are often embedded in laptops and desktops to manage hardware encryption, including features in systems like BitLocker.

Hybrid Encryption

Hybrid encryption combines elements of both software and hardware encryption.
While it is primarily software-based, it can be linked to a specific hard drive and often utilizes the TPM chip for enhanced security. This means that hybrid encryption can provide faster performance than traditional software encryption by leveraging the capabilities of the hardware.
Access methods for hybrid encryption can vary:

- transparent mode, where the encryption process is seamless;
- user authentication, where a password or PIN is required;
- USB authentication, which requires a USB device to access the data.

A well-known example of hybrid encryption is **Windows BitLocker**, which provides robust encryption options while maintaining ease of use.

How to circumvet HW encryption?

As for software encription, we can resort to the following strategies:

1. **Cold reboot attack**: a cold reboot attack targets the system’s memory, specifically DRAM (Dynamic Random Access Memory). When a computer is powered down and restarted quickly, the data in DRAM may not be fully erased. An attacker can exploit this by booting the computer with a malicious operating system from an external source, like a USB drive. This method allows the attacker to access and copy the contents of the DRAM, including encryption keys, before the memory is fully cleared. With these keys, they can decrypt the hard disk and access the stored data.

2. **Data outside encrypted area**: pieces of files may be found outside the encrypted area. This happens when files are stored temporarily or cached by the operating system, creating unprotected areas on the disk. These unencrypted parts can provide useful information and clues that help recover or reconstruct sensitive data.

3. **OS and password integration**: depending on how the encryption is set up, incorrect password attempts may still allow partial access to the system's memory or files. For example, some encryption programs may temporarily load parts of the operating system or application files into memory before fully verifying the password. If an attacker can access these areas, they may be able to gather enough information to bypass the encryption.

As for the strategies related to hardware encription, we can remind:

1. **Data outside the encrypted area**: similarly to software encryption, hardware encryption can still leave traces of data outside the protected zone. If temporary files or system logs are not encrypted, an attacker may be able to access fragments of information that could help reconstruct the encrypted data.
2. **Avoid sleep and shutdown states**: hardware encryption systems are most vulnerable during sleep or shutdown states. When a device is put to sleep or powered down, it requires re-entry of the password when it is restarted. An attacker might attempt to exploit this by using cold boot techniques before the system fully shuts down. By accessing the system during these transitions, they may gain unauthorized access to encryption keys or sensitive data.
3. **Recovery keys stored in clear view**: some encryption systems create recovery keys as a backup option for users who forget their passwords. Unfortunately, if these recovery keys are stored in clear text or unprotected areas, they become a security risk. An attacker who gains access to these keys can easily decrypt the entire disk without needing the original password.
4. **Exploiting TPM weaknesses**: the TPM chip is crucial for managing encryption keys in hardware encryption systems. If these are stored insecurely or if TPM itself has some bugs, keys could be accessed directly by an attacker.

Despite these vulnerabilities, encryption remains an effective tool for protecting data and, if necessary, destroying evidence. If encryption keys are reset or wiped, all data on the hard disk becomes unreadable and irretrievable.

Mobile forensics

Mobile forensics is the process of recovering and analyzing data from mobile devices, such as smartphones and tablets, for investigative purposes. Unlike traditional computer forensics, mobile forensics goes beyond just finding files like documents or pictures. It involves looking for a variety of digital traces left by the device and its usage.

When examining mobile devices, experts look for many different types of information, including:

- **IMEI code (International Mobile Equipment Identifier):** a special 15-digit number that is unique to every mobile device. This code helps to identify and track the phone within the mobile network. The IMEI code provides important details such as the vendor (who made the phone), the model (what kind of phone it is), and even the country where it was made or sold. Mobile networks use this code to manage

devices, verify their identity, and sometimes to block stolen phones from accessing the network.

IMEI code can be retrieved:

- look on the back or under the battery (switched off devices)
- dialling *#06# (switched on devices – beware of alterations)

- SIM **(Subscriber Identity Module) card:** it is needed to connect a phone to a mobile network. It stores important information that allows the phone to make calls, send messages, and use mobile data. Each SIM card has its own unique code called the *Integrated Circuit Card Identification (ICCID)* number, which is used to identify the SIM card within the network. This number helps mobile providers manage and authenticate the services linked to each SIM card.

- **Images and media files**: photos, videos, and other media stored on the device can be analyzed to understand the user's activities, locations, and contacts. Metadata from images can also reveal when and where they were taken.

Seizure/collection of a mobile device

This process involves several steps aimed at securing the device, documenting its state, and minimizing any potential interference. As a matter of fact, it is possible to carrying on this process by the following list of stages.

1. **Secure the phone/tablet:** device's evidences must be physically protected. This means taking possession of the device as quickly as possible to prevent any tampering, deletion of data, or loss of information. The device should be kept under the direct supervision of the personnel conducting the seizure to ensure its safety.

2. **Isolate the device and prevent unauthorized access:** the device must be isolated from any wireless network, e.g., bluetooth connections, WiFi or other external devices. Several solutions can achieve this (e.g., Faraday bag, airplane mode). The aim is to prevent remote access, synchronization, or deletion of data through network connections. It is important to ensure that no unauthorized personnel or third parties operate or interact with the device during this stage.

3. **Note any physical defects:** observing and recording any physical defects or abnormalities on the device is a crucial part of the seizure process. This may include checking for a broken display, dents, scratches, or other signs of damage. These observations should be clearly noted in the documentation to provide a full account of the device's condition at the time of seizure. It is essential for this documentation to be as detailed as possible, as it may be used for reference in future examinations or as part of an evidence record.

4. **Take photographs of the device:** It is necessary to take photographs of the device to visually document its state. These photographs serve as visual evidence and can support any written notes regarding physical defects. Ensure that the images are clear, properly focused, and show all identifying features of the device, such as serial

numbers, logos, or other relevant markings. The photographs should be time-stamped and securely stored in the case file.

5. **Document all actions taken:** Throughout the entire seizure process, it is vital to keep a detailed log of every operation. This includes recording the time and place of the seizure, the condition of the device, the names of personnel involved, and the steps taken to secure and isolate the device. Accurate documentation is essential for maintaining the chain of custody, ensuring transparency, and supporting the integrity of the evidence in legal proceedings.

6. **Verify if the device is on or off:** After securing the device, it is important to check whether it is powered on or off. This step should be performed carefully to avoid accidentally altering or accessing any data on the device. If the device is found to be on, note its status in the documentation. If the screen is locked, do not attempt to unlock it as this could alter important timestamps or other forensic evidence.

If the device is off, leave it switched off.
Powering on the device can cause it to connect to networks, sync with cloud services, or overwrite crucial data, compromising its integrity as evidence. Instead, leaving the device off also helps preserve its current state, allowing forensic experts to extract data under controlled conditions later.

If the device is on, secure data.
Following a proper isolation procedures preserve the integrity of the data and prevent any unauthorized access or modifications. The following steps outline the process for effective isolation:

1. Document the information on the screen: the first action is to carefully observe and document any displayed information including notifications, active applications, call logs, messages, or any other visible data. Photographs can be taken to capture the screen exactly as it appears, ensuring that evidence is preserved for later analysis. This is critical because any displayed information might provide key insights, such as recent activities or app usage.

2. Register the date and time
This information is needed to cross-check the actual time to identify any potential discrepancies or offsets. This timestamp information could later be used to establish timelines or sequences of events, making it a vital piece of data.

3. Avoid navigating through the device
To maintain the integrity of the evidence, it is essential to avoid interacting with the device beyond the initial observations (e.g., navigating through menus, opening messages, or interacting with any apps, …). Even seemingly harmless interactions could compromise valuable evidence, so the device should remain untouched until forensic experts can analyze it in a controlled environment.

4. Isolate the device from network connections
In order to prevent any external access or manipulation of data, the device must be isolated to prevent any connections to networks or other device by means of:

- bluetooth (2.45 GHz)
- Wi-Fi
- Cellular Networks (GSM/UMTS/LTE
- GPS (1575 MHz and 1227 MHz)[1]

[1] it could update location data or be used for tracking.

To achieve this level of isolation, we can resort to a **Faraday bag,** a protective enclosure blocks electromagnetic signals, effectively preventing the device from connecting to any networks while keeping it in its current state. Maintaining the phone in a Faraday bag ensures that the device remains powered on, preserving its state without allowing any external interference until further forensic analysis is possible.
Another approach is using a **jammer**, which emits signals that block specific frequency ranges, preventing the device from connecting. Jammers can be effective, but they have limitations like requiring power to operate and being restricted or illegal in some regions.

Note that jammers or Faraday's cages imply a waste of battery (the phone keeps on trying to connect); therefore, you need for external power charge to keep it on.

The third method involves enabling **airplane mode** on the device, i.e., disabling most wireless connections. However, investigators must manually verify that all connections are truly off since some devices may allow certain connections to remain active even in airplane mode. This method is quick and easy but requires direct interaction with the device, which could potentially alter data (if you do not know the model you still can mess up the things ....)

5. **Switch it off** with normal shutdown or removing the battery. Remember that switching the device off could enable phone lock and require an unlocking code (e.g., PIN of SIM card or lock number of the phone).

When collecting the phone, it could be useful to take other things with you: collect with the phone:

- Cables
- Battery charger
- Wrappings
- Memories
- Manuals
- CDs, USB pens storing software suites to manage the phone
- Phone bills related to the phone
- SIM envelope with PIN, PUK codes

Document the seizure with the following data:

- Operator's name
- Date and time of the seizure
- Where the device has been taken (address, GPS position, etc.)

Data carving on a mobile device

Considering a mobile device, data storage can take place at different components.

- **SIM card** is essential for accessing mobile networks and typically holds user identification, phone numbers, and basic contact information.
- **SD cards** (Secure Digital cards), allow users to expand their storage capacity for photos, videos, and apps.
- **integrated memory**, often referred to as internal storage, is where the operating system, applications, and data are located.

Integrated Memory.
The integrated memory of a device can be accessed by means of two acquisition strategies:

- Logic acquisition (files)
- Physical acquisition: cloning NAND memory

Logic acquisition can be done using

- Specific softwares[2]
- hardware devices.

Physical acquisition can be done with different tools; note that it depends on the specific device and OS.

SIM cards.

Retrieving data from SIM cards can be tricky because SIMs can be locked and the information stored on them is often protected with passcode numbers. To access a locked SIM card, you need to unlock the phone using a **PIN code** (Personal Identification Number), which is usually a four-digit number. You have three chances to enter the correct PIN. If you make three mistakes, you will need to use a **PUK** (Personal Unblocking Key), which is a ten-digit code that allows you ten more tries to unlock the SIM.

The data on a SIM card is organized in a tree structure, which helps in managing and finding the stored information efficiently. Several folders can be found containng different types of information such as messages, contacts, IMEI or SIM cards.

[2] *Need for a reliable connection between mobile device and computer such as cables, infrared, bluetooth (problem: infrared and Bluetooth could change the state of device during the connection- using cables is better)*